

Java Variables: Real-Life Use Cases

Seeing variables in action with practical code examples

Programming is about solving problems. These examples show how we use data types to represent real-world objects and math.

1. Student Profile

```
String name = "John Doe";  
int id = 15;  
float fee = 75.25f;  
char grade = 'B';  
  
System.out.println("ID: " + id);
```

Data Mix:

Shows how String, int, float, and char work together to build a complete record of a person.

2. Rectangle Area Calculator

```
int length = 4, width = 6;  
int area = length * width;  
System.out.println("Area: " + area);
```

Processing Data:

Variables aren't just for storage; they are used in **expressions** to create new information.

3. Temperature Converter (C → F)

```
double celsius = 25.0;  
double fahr = (celsius * 9/5) + 32;  
System.out.println(fahr + "°F");
```

Precision:

Uses double to handle precise decimal calculations required for scientific or weather data.

4. Swapping Two Numbers (Temp Variable)

```
int a = 5, b = 10, temp;  
temp = a; // temp is 5  
a = b; // a is 10  
b = temp; // b is 5
```

The "Third Cup" Logic:

To swap two values, you need a temporary container (temp) so the first value isn't lost.

5. Constant-Based Math (Circle Area)

```
final double PI = 3.14159;  
double radius = 7.0;  
double area = PI * radius * radius;
```

Safety with final:

By making PI a constant, we ensure that the value of Pi cannot be changed accidentally by other code.

6. Shopping Cart Logic

```
double price = 2.50;  
int qty = 4;  
double total = price * qty;  
System.out.println("Bill: $" + total);
```

Financial Basics:

Multiplying price (decimal) by quantity (whole number) results in a total cost.

Key Takeaways

- Variables store **real-world attributes**.
- Calculations turn input into **output**.
- `double` is preferred for decimals in math.
- Use temp variables for logic swaps.
- `final` protects fixed mathematical values.
- Concatenation (+) helps build readable reports.